

ANICHECK: A MOBILE APPLICATION FOR RICE LEAF DISEASE IDENTIFICATION USING CONVOLUTIONAL NEURAL NETWORKS FOR SMALLHOLDER FARMERS IN BARANGAY MASAGANA, NUEVA ECIJA

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This document is a synthetic sample created for software demonstration purposes. The institution, authors, location, dataset, and all reported results are fictional.

ABSTRACT

Rice is the primary staple crop of the Philippines, and leaf diseases remain among the leading causes of yield loss among smallholder farmers. Accurate diagnosis typically requires an agricultural extension technician, but the ratio of technicians to farming households in many rural municipalities makes timely consultation impractical. Farmers therefore rely on visual judgment and informal advice, which frequently results in the application of the wrong pesticide or the wrong dosage.

This study developed AniCheck, an Android application that identifies four common rice leaf diseases — bacterial leaf blight, brown spot, leaf smut, and tungro — as well as healthy leaves, from a photograph captured on the farmer's own phone. The system uses a MobileNetV2 convolutional neural network fine-tuned through transfer learning on a dataset of 3,240 leaf images collected by the proponents in Barangay Masagana, Nueva Ecija, over a six-week period. The trained model was converted to TensorFlow Lite and embedded in the application so that inference runs on the device without requiring an internet connection.

The model achieved an overall classification accuracy of 94.7% on a held-out test set of 648 images. The application was then evaluated by 30 respondents drawn from the same barangay using an instrument adapted from the ISO/IEC 25010 software quality model, covering functional suitability, usability, reliability, and portability. The respondents rated the system with an overall weighted mean of 4.31, interpreted as Very Acceptable. The proponents conclude that on-device disease identification is a viable form of decision support for smallholder farmers and recommend expansion of the disease coverage and a longer field validation period.

Keywords: rice leaf disease, convolutional neural network, transfer learning, MobileNetV2, mobile application, precision agriculture, ISO/IEC 25010

CHAPTER 1

INTRODUCTION

Background of the Study

Agriculture contributes roughly a tenth of Philippine gross domestic product, and rice occupies the largest share of cultivated land in Central Luzon. Nueva Ecija alone accounts for a substantial fraction of national palay production, and the province is commonly described as the rice granary of the Philippines. Within this context, the economic exposure of an individual smallholder to a single failed season is severe: most operate on landholdings of under two hectares and finance inputs through seasonal credit.

Leaf diseases are a recurring source of that exposure. Bacterial leaf blight, brown spot, leaf smut, and tungro each present through discoloration and lesion patterns on the leaf surface, and each demands a different management response. Bacterial leaf blight is bacterial and does not respond to fungicide; brown spot is fungal and frequently signals an underlying potassium deficiency; tungro is viral and transmitted by the green leafhopper, meaning that treatment of the plant itself is futile and control must target the vector. Applying a fungicide to a tungro infection therefore costs the farmer the price of the chemical, the labor of application, and the time during which the actual infection continues to spread.

The Department of Agriculture maintains an agricultural extension service intended to provide exactly this diagnostic support. In practice, the proponents' initial interviews in Barangay Masagana indicated that farmers consulted an extension technician a median of once per cropping season, and that the most common source of diagnostic advice was a neighboring farmer or the sales staff of the local agricultural supply store. The latter is a source with an evident commercial interest in the recommendation.

Advances in computer vision offer a plausible intervention. Convolutional neural networks have demonstrated strong performance on plant disease classification benchmarks, and architectures designed for constrained hardware now make it feasible to run inference directly on a mid-range smartphone. Since smartphone ownership among farming households in the barangay was observed to be high while reliable mobile data coverage was not, an application that performs classification on the device rather than on a remote server is the more appropriate deployment model.

Statement of the Problem

The study sought to answer the following questions:

1. What are the current practices of smallholder rice farmers in Barangay Masagana with respect to identifying and responding to leaf disease?
2. How may a mobile application be designed and developed to identify rice leaf diseases from a photograph captured on a farmer's own device?

3. What level of classification accuracy does the trained convolutional neural network achieve on a held-out test set?
4. How acceptable is the developed application to its intended users when evaluated against the ISO/IEC 25010 characteristics of functional suitability, usability, reliability, and portability?

Objectives of the Study

The general objective was to develop and evaluate a mobile application capable of identifying common rice leaf diseases without requiring an internet connection. Specifically, the study aimed to assemble a labeled image dataset of rice leaves representing four disease classes and a healthy class; to train a convolutional neural network on that dataset using transfer learning; to convert and embed the trained model in an Android application with an interface suitable for users with limited formal schooling; and to evaluate both the classification performance of the model and the acceptability of the application to its intended users.

Scope and Limitations

The application identifies four disease classes — bacterial leaf blight, brown spot, leaf smut, and tungro — plus a healthy class. These four were selected because they were the diseases most frequently named by farmers during the initial interviews and because they present visible foliar symptoms. Diseases whose symptoms appear principally on the panicle, stem, or root, including sheath blight and false smut, are outside the coverage of the system.

The image dataset was collected exclusively within Barangay Masagana during a six-week window in the 2024 wet season. The application runs on Android 9.0 and above; no iOS version was produced. The system returns an identification and a confidence percentage; it does not recommend a specific chemical product or dosage, since the proponents judged that prescribing agricultural chemicals falls outside both the competence of the development team and the intended function of a diagnostic aid. The application is intended as a decision-support tool and not as a replacement for consultation with a qualified agricultural technician.

Significance of the Study

For smallholder farmers, the study offers a means of obtaining a preliminary disease identification at the moment of observation rather than at the next available consultation. For agricultural extension officers, a tool that handles routine identification may allow scarce technician time to be directed toward the cases that genuinely require expert judgment. For the local government unit, the aggregate pattern of identifications could in principle inform the targeting of advisory campaigns, although the present version of the system does not transmit or aggregate any data. For future researchers, the study contributes a documented dataset collection procedure and a baseline set of results against which alternative architectures may be compared.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Convolutional Neural Networks in Plant Pathology

The application of convolutional neural networks to plant disease classification was established at scale by Mohanty, Hughes, and Salathé (2016), who trained AlexNet and GoogLeNet on the PlantVillage dataset of 54,306 images spanning 14 crop species and 26 diseases, reporting classification accuracy above 99% on held-out data. The authors themselves noted a significant qualification: performance degraded sharply when the trained models were evaluated on images taken under conditions different from those of the training set, since the PlantVillage images were captured against uniform backgrounds under controlled lighting.

Ferentinos (2018) extended this line of work across a larger set of architectures and reported comparable results, again on laboratory-condition imagery. Barbedo (2018) examined the limitations of the approach directly and argued that the dominant constraint on real-world plant disease classification is not model architecture but the size and diversity of available datasets, observing that models trained on images of individual detached leaves frequently fail on photographs of plants in the field.

Transfer Learning and Lightweight Architectures

Training a deep convolutional network from random initialization requires a volume of labeled data that is rarely available in agricultural settings. Transfer learning addresses this by initializing the network with weights learned on a large general dataset, typically ImageNet, and fine-tuning only the later layers on the target task. Yosinski, Clune, Bengio, and Lipson (2014) demonstrated that features learned in the early layers of a convolutional network are largely generic across visual domains, which provides the theoretical basis for the technique.

Howard et al. (2017) introduced the MobileNet family, which replaces standard convolutions with depthwise separable convolutions and thereby reduces both parameter count and computation substantially at a modest cost in accuracy. Sandler et al. (2018) refined the architecture with inverted residual blocks and linear bottlenecks in MobileNetV2. These architectures are the standard choice where inference must run on a mobile device rather than a server.

Technology Acceptance in Agricultural Contexts

Technical performance does not by itself determine whether a tool is adopted. The Technology Acceptance Model of Davis (1989) holds that adoption is governed principally by perceived usefulness and perceived ease of use. Venkatesh, Morris, Davis, and Davis (2003) later consolidated competing models into the Unified Theory of Acceptance and Use of Technology, adding social influence and facilitating conditions as determinants. In agricultural extension specifically, Rogers' (2003) diffusion of innovations framework remains widely applied, with its emphasis on trialability

and observability as properties that accelerate adoption among farming populations.

Related Systems

Several mobile applications addressing plant disease identification have been documented. Plantix, developed by PEAT GmbH, covers a broad range of crops and diseases but performs classification on a remote server and therefore requires connectivity at the time of use. Nuru, developed within the PlantVillage project, operates offline and targets cassava disease in East Africa, demonstrating that on-device inference is workable in low-connectivity agricultural settings. Locally, Ramos and Tolentino (2019) developed a rice disease identification prototype for Philippine conditions, though their system required a server connection and their evaluation was confined to laboratory testing without farmer respondents.

Synthesis and Research Gap

The reviewed literature establishes that convolutional neural networks classify plant leaf disease with high accuracy under controlled conditions, that transfer learning makes such models trainable on modest datasets, and that lightweight architectures make on-device inference feasible. Two gaps remain relevant to the present study. First, the majority of reported systems either depend on server-side inference or were evaluated without the participation of the farmers for whom they were intended. Second, published Philippine work on rice leaf disease identification has generally stopped at the prototype stage without an acceptability evaluation conducted with end users. The present study addresses both by developing an offline system and evaluating it with respondents drawn from the intended user population.

CHAPTER 3

METHODOLOGY

Research Design

The study employed a developmental research design. The developmental component covered the construction of the dataset, the training of the classification model, and the implementation of the mobile application. The descriptive component covered the evaluation of the completed system by respondents using a structured survey instrument. Software development followed an iterative, agile approach organized into four two-week sprints, with a working build produced at the end of each sprint and reviewed with the adviser.

Dataset Collection

Leaf images were collected from eleven rice fields within Barangay Masagana over a six-week period spanning the vegetative and reproductive stages of the 2024 wet season crop. Photographs were captured using a Samsung Galaxy A54 at a resolution of 3000 by 4000 pixels. To ensure consistency, each leaf was detached and photographed flat against a white cardboard background under natural daylight between 8:00 AM and 10:00 AM, with the camera held approximately 25 centimeters from the leaf surface.

Labeling was performed by an agricultural technician from the municipal agriculture office, who identified each photographed leaf on site. Images that the technician could not confidently classify were discarded. A total of 3,240 images were retained. The class distribution is presented in Table 1.

Class	Images	Share of Dataset
Bacterial Leaf Blight	624	19.3%
Brown Spot	718	22.2%
Leaf Smut	586	18.1%
Tungro	412	12.7%
Healthy	900	27.8%
Total	3,240	100.0%

Table 1. Class distribution of the collected image dataset.

The tungro class is visibly underrepresented relative to the others. This reflects the actual incidence encountered during the collection window rather than a sampling decision. To compensate, data augmentation was applied more aggressively to this class during training.

Preprocessing and Augmentation

All images were resized to 224 by 224 pixels to match the input dimensions expected by MobileNetV2, and pixel values were normalized to the range expected by the pretrained weights. The dataset was partitioned into a training set and a test set using an 80:20 split, yielding 2,592 training images and 648 test images. The split was stratified so that class proportions were preserved in both partitions.

Augmentation was applied to the training partition only, comprising random rotation within plus or minus 25 degrees, horizontal and vertical flipping, random zoom within 15%, and brightness adjustment within 20%. Augmentation was applied online during training rather than by generating and storing additional image files.

Model Architecture and Training

MobileNetV2 pretrained on ImageNet was selected as the base network. The original classification head was removed and replaced with a global average pooling layer, a dropout layer with rate 0.3, and a dense softmax layer of five units corresponding to the five classes. The convolutional base was frozen for the first 15 epochs so that only the new head was trained, after which the final 30 layers of the base were unfrozen and the entire network was fine-tuned at a reduced learning rate for a further 25 epochs.

Training used the Adam optimizer with an initial learning rate of 0.001, reduced to 0.0001 for the fine-tuning phase, and categorical cross-entropy as the loss function. Batch size was set to 32. Training was performed on the free tier of Google Colaboratory using an NVIDIA T4 GPU. Several combinations of dropout rate, learning rate, and number of unfrozen layers were tried, and the configuration reported here is the one that produced the highest accuracy on the test set.

The trained model was converted to TensorFlow Lite with post-training dynamic range quantization, reducing the model file from 14.2 MB to 3.8 MB. The quantized model was embedded directly in the application package.

System Architecture and Development Tools

The application was implemented in Kotlin using Android Studio Hedgehog, targeting Android API level 28 and above. CameraX handles image capture, and the TensorFlow Lite Android interpreter performs inference. A local Room database stores the user's identification history on the device. No backend server, remote database, or network call is present in the application; the entire pipeline from capture through inference to display executes locally.

The interface was deliberately constrained to three screens: a camera screen, a result screen showing the identified class with a confidence percentage and a short description of the disease in Filipino, and a history screen. Text was kept minimal and each disease description was paired with a reference photograph, on the reasoning that a largely text-based interface would be poorly suited to the intended users.

Evaluation Design

Model performance was evaluated on the 648-image test set using accuracy, precision, recall, and F1-score computed per class, together with a confusion matrix.

System acceptability was evaluated using a survey instrument adapted from ISO/IEC 25010, covering four characteristics: functional suitability, usability, reliability, and portability. The instrument consisted of twenty statements, five per characteristic, rated on a five-point Likert scale. The instrument was translated into Filipino and reviewed for clarity by the adviser and by one faculty member from the college.

Thirty respondents were selected through purposive sampling from among the rice farmers of Barangay Masagana, the criteria being active cultivation of rice during the current season and ownership of an Android smartphone. Each respondent attended a single session in which the application was installed on their device, its use was demonstrated, and they were guided through capturing between three and five identifications on leaves available at the session. Sessions ran approximately thirty minutes. The survey was administered immediately afterward, with the proponents reading each statement aloud and recording the response.

Responses were analyzed using weighted mean. The scale was interpreted according to the ranges in Table 2.

Range	Verbal Interpretation
4.21 - 5.00	Very Acceptable
3.41 - 4.20	Acceptable
2.61 - 3.40	Moderately Acceptable
1.81 - 2.60	Slightly Acceptable
1.00 - 1.80	Not Acceptable

Table 2. Likert scale interpretation ranges.

Ethical Considerations

Permission to conduct the study within the barangay was obtained in writing from the barangay captain. Participation was voluntary and respondents were informed that they could withdraw at any point. Consent was obtained verbally at the start of each session, since several respondents indicated discomfort with signing documents. No personally identifying information was recorded on the survey forms, and the application stores identification history only on the user's own device.

CHAPTER 4

RESULTS AND DISCUSSION

Current Farmer Practices

The initial interviews conducted with 24 farmers established the baseline the system was intended to improve upon. Nineteen of the 24 reported that their usual response to observing an unfamiliar leaf symptom was to consult a neighboring farmer, and 16 reported purchasing a pesticide on the recommendation of the agricultural supply store. Eight reported having applied a treatment that they later concluded had been inappropriate for the actual problem. Median reported consultation with an extension technician was once per cropping season.

Model Classification Performance

The trained model achieved an overall accuracy of 94.7% on the 648-image test set. Per-class performance is presented in Table 3.

Class	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	0.892	0.813	0.851	125
Brown Spot	0.941	0.958	0.949	144
Leaf Smut	0.956	0.949	0.952	117
Tungro	0.928	0.939	0.933	82
Healthy	0.978	0.994	0.986	180
Weighted Average	0.946	0.947	0.946	648

Table 3. Per-class classification performance on the test set.

Performance is strongest on the healthy class, where recall reaches 0.994. The weakest result is the recall of 0.813 on bacterial leaf blight, indicating that approximately one in five bacterial leaf blight samples in the test set was assigned to another class. The confusion matrix in Table 4 shows that the majority of these misclassifications were assigned to brown spot.

Actual \ Predicted	BLB	BS	LS	TU	HL
Bacterial Leaf Blight	102	16	5	2	0
Brown Spot	5	138	1	0	0
Leaf Smut	3	2	111	1	0
Tungro	3	1	0	77	1
Healthy	1	0	0	0	179

Table 4. Confusion matrix on the 648-image test set.

The confusion between bacterial leaf blight and brown spot is consistent with the visual similarity of the two conditions in their early stages, where both present as yellowing along the leaf margin before the characteristic lesion patterns become distinct. This is also the confusion with the greatest practical cost, since the two diseases call for different treatment classes.

On-Device Inference Performance

Inference latency was measured on three devices, presented in Table 5. Each figure is the mean of twenty consecutive classifications.

Device	RAM	Mean Inference Time
Samsung Galaxy A54	8 GB	312 ms
Xiaomi Redmi 12	4 GB	487 ms
Samsung Galaxy A12	3 GB	894 ms

Table 5. Mean inference latency by device.

Latency remained under one second on all three devices, which the proponents considered acceptable for the intended interaction pattern. The oldest device tested carried 3 GB of RAM; behavior on devices below this specification was not established.

System Acceptability

The results of the ISO/IEC 25010 evaluation with the 30 respondents are summarized in Table 6.

Characteristic	Weighted Mean	Interpretation
Functional Suitability	4.38	Very Acceptable
Usability	4.42	Very Acceptable
Reliability	4.09	Acceptable
Portability	4.35	Very Acceptable
Overall	4.31	Very Acceptable

Table 6. Summary of respondent evaluation by ISO/IEC 25010 characteristic.

Usability received the highest rating at 4.42. During the sessions the proponents observed that respondents were able to complete a full capture-and-identify cycle without assistance after a single demonstration, which supports the decision to constrain the interface to three screens and to minimize text.

Reliability received the lowest rating at 4.09, the only characteristic falling below the Very Acceptable threshold. The comments recorded during these sessions indicate that the rating was driven principally by instances in which the application returned an identification that the

respondent believed to be incorrect. In several such instances the leaf in question carried symptoms of more than one condition, a case the present system cannot express since it returns a single class.

Discussion

The overall accuracy of 94.7% is lower than the figures commonly reported in the plant disease classification literature, where results above 99% on the PlantVillage dataset are routine. The proponents attribute the difference principally to dataset scale: 3,240 images across five classes is an order of magnitude smaller than the datasets on which those figures were obtained.

The acceptability results indicate that the intended users found the system usable and useful. The gap between the usability rating and the reliability rating is instructive: respondents were more confident in their ability to operate the application than in the correctness of what it told them. For a diagnostic tool this is the more consequential of the two dimensions, and it corresponds directly to the per-class weakness identified on bacterial leaf blight.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

Summary of Findings

The study established that farmers in Barangay Masagana rely predominantly on informal sources for disease diagnosis and that misapplied treatment is a recurring outcome. A MobileNetV2 model fine-tuned on 3,240 locally collected images achieved 94.7% accuracy on a held-out test set, with the weakest per-class recall of 0.813 on bacterial leaf blight. The model was successfully deployed for on-device inference with latency under one second on all three test devices. Thirty respondents rated the completed application with an overall weighted mean of 4.31, interpreted as Very Acceptable, with reliability rated lowest at 4.09.

Conclusion

The proponents conclude that on-device convolutional neural network inference is a technically viable approach to rice leaf disease identification under the connectivity and hardware conditions prevailing in the study area, and that the resulting application is acceptable to its intended users. The system is best characterized as a preliminary decision-support aid. The measured per-class performance on bacterial leaf blight, together with the reliability rating, indicates that it should not at present be treated as a substitute for consultation with an agricultural technician in cases where a costly treatment decision follows from the identification.

Limitations

The dataset was collected within a single barangay over a six-week window, and every training image was captured from a detached leaf against a uniform background under morning daylight. Application in the field involves photographs of attached leaves against complex backgrounds under variable light. The extent to which measured test accuracy transfers to that setting was not established by this study.

The evaluation was conducted with 30 respondents from the same barangay in which the dataset was collected, immediately following a guided demonstration session. It therefore measures initial impression rather than sustained use, and the respondents are not independent of the population whose fields supplied the training images.

The system recognizes five classes and always returns one of them together with a confidence percentage. Behavior on a leaf affected by a condition outside the five classes, on a leaf affected by two conditions simultaneously, or on a photograph that is not of a rice leaf at all, was not systematically characterized.

Recommendations

The proponents recommend, first, that the dataset be expanded to include images captured in the field under the conditions of actual use, and that images be collected from multiple municipalities across more than one cropping season. Second, that the disease coverage be extended to include sheath blight and false smut, and that a mechanism be introduced for expressing multiple simultaneous conditions or an inconclusive result rather than forcing a single class. Third, that a validation study be conducted in which the application's identifications are compared against the judgment of an agricultural technician on the same leaves, so that accuracy may be assessed against the standard the tool is intended to approximate rather than against a held-out partition of its own dataset. Fourth, that a longitudinal evaluation be conducted with farmers using the application unsupervised across a full cropping season. Fifth, that an iOS version be developed to extend coverage beyond Android users.

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